Quantification: 699_HYP_HP_z11c1-2
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PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
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measure	value
lacuna count	19
interior lacuna count	16
frame-edge lacuna count	3

Interior lacunae only: mean, median and sample SD (n = 16)
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measure	kind	mean	median	sd	units
area_px2	lacuna shape	2731.81	1654	3092.58	px^2
major_axis_length_px	lacuna shape	88.7386	70.9441	53.7147	px
minor_axis_length_px	lacuna shape	43.4742	35.3308	20.5257	px
aspect_ratio	lacuna shape	2.1103	1.7031	0.9618	unitless
eccentricity	lacuna shape	0.7914	0.8095	0.1525	unitless
solidity	lacuna shape	0.7086	0.6984	0.0908	unitless
roots_count	headline	10.3125	11.5	5.0559	count
ring_length_r30_px	headline	396.25	381.5	213.056	px
ring_length_r60_px	reported	728.5	656.5	423.552	px
owned_length_px	ownership-dependent	931.868	805.769	805.743	px
edge_count	ownership-dependent	50.5	36	46.2645	count
mean_edge_length_px	ownership-dependent	19.6204	20.0113	2.6404	px
perimeter_px	normalised	347.628	269.513	225.354	px
ring_area_r30_px2	normalised	9845.94	8062.5	4697.34	px^2
ring_area_r60_px2	normalised	22082.6	19536	9877.06	px^2
in_frame_fraction_r30	normalised	0.9948	1	0.0209	unitless
in_frame_fraction_r60	normalised	0.9892	1	0.0432	unitless
ring_density_r30	normalised	0.041669	0.0437626	0.0158225	px^-1
ring_density_r60	normalised	0.0331852	0.0336134	0.0117621	px^-1
roots_per_100px_perimeter	normalised	3.4044	3.4676	1.7864	per 100 px
ring_attached_length_r30_px	network v2	349.938	319	220.224	px
ring_attached_length_r60_px	network v2	558.75	481.5	408.973	px
ring_length_w_r30_px	network v2	446.195	433.519	238.684	px
ring_length_w_r60_px	network v2	817.214	734.857	472.722	px
sholl_crossings_r10	network v2	9.375	8	5.7721	count
sholl_crossings_r20	network v2	9.5	10	4.487	count
sholl_crossings_r30	network v2	9.8125	9	5.1279	count
loop_count	loops, report only	2.875	0.5	4.3493	count
loop_edge_share	loops, report only	0.1231	0.0469	0.1434	share
ring_width_mean_r30_px	width	7.4811	7.7238	1.0638	px

Per field
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measure	value	units
analysed_area_px2	825799	px^2
total_skeleton_length_px	19679	px
canalicular_length_density_per_px	0.0238303	px^-1
junction_count	433	count
junction_density_per_px2	0.00052434	px^-2
lacuna_count	19	count

lacunae_per_px2	2.301e-05	px^-2
skeleton_component_count	285	count
mean_component_length_px	69.0491	px
median_component_length_px	15	px
field_length_density_w_per_px	0.0266057	px^-1
field_density_without_flagged_per_px	0.0182211	px^-1
field_density_in_roi_per_px		px^-1
unattached_length_px	563	px
unattached_component_count	29	count
canalicular_length_density_attached_per_px	0.0238303	px^-1
field_length_density_with_unattached_per_px	0.024512	px^-1
excluded_area_px2	161365	px^2
analysed_field_area_px2	887211	px^2
loop_count	62	count
loop_edge_share	0.1922	share
edge_count	1301	count

Canalicular width, the new measure

measure	value	units
width_mean_px	7.6113	px
width_median_px	8	px
width_p10_px	4	px
width_p90_px	10.7703	px
width_pixels_used	17309	count

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel.
Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the
dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the
pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every
thread), and it has not been validated against any manual measurement.

Lacuna shape

lacuna_id	area_px2	major_axis_length_px	minor_axis_length_px	aspect_ratio	eccentricity	solidity	orientation_rad	centroid_row_px	centroid_col_px	on_border
1	483	34.5143	18.5791	1.8577	0.8428	0.9028	-1.3729	9.6004	721.977	True
2	971	48.2689	39.9508	1.2082	0.5612	0.6637	-1.1736	126.652	613.53	False
3	1293	67.1123	29.3143	2.2894	0.8996	0.7085	-0.9693	147.124	738.418	False
4	933	52.1712	30.6068	1.7046	0.8098	0.7272	-0.9318	146.222	325.69	False
5	8385	126.932	101.718	1.2479	0.5982	0.7052	-1.4611	277.754	204.132	False
6	1905	74.7758	58.1104	1.2868	0.6293	0.5641	1.396	301.221	621.288	False
7	1403	54.0186	34.896	1.548	0.7633	0.888	-0.1596	368.566	862.24	False
8	3093	92.0714	54.1062	1.7017	0.8091	0.6916	1.2856	410.259	393.304	False
9	1944	82.0911	39.6724	2.0692	0.8755	0.6506	1.2002	562.526	659.101	False
10	1195	44.9336	38.8022	1.158	0.5043	0.7607	-1.4455	562.059	755.791	False
11	2018	112.423	30.231	3.7188	0.9632	0.6731	1.4432	571.53	456.488	False
12	634	48.158	30.1352	1.5981	0.78	0.5992	-1.5103	631.199	551.279	False
13	1329	62.4213	28.6247	2.1807	0.8887	0.8819	1.2484	643.479	606.14	False
14	812	43.7603	30.951	1.4139	0.7069	0.7167	-1.4386	655.233	193.442	False
15	12055	251.462	78.5981	3.1993	0.9499	0.6372	-0.5972	767.904	745.585	False
16	2798	118.207	35.7656	3.305	0.9531	0.8127	-1.3169	814.447	954.982	False
17	2396	80.3638	41.3693	1.9426	0.8573	0.8812	-1.0185	852.972	994.821	True
18	14977	374.03	83.4021	4.4847	0.9748	0.5035	-1.4304	910.923	184.857	True
19	2941	141.011	34.1041	4.1347	0.9703	0.6575	1.2599	895.57	423.196	False

Network per lacuna

lacuna_id	roots_count	ring_length_r30_px	ring_length_r60_px	owned_length_px	edge_count	mean_edge_length_px	ring_width_mean_r30_px
1	3	114	255	112.598	8	14.0747	5.9307
2	12	377	731	915.181	53	17.2676	8.4862
3	12	386	697	821.826	35	23.4807	8.1517
4	8	302	589	854.311	54	15.8206	9.3181
5	18	887	1678	2269.63	131	17.3254	7.7688
6	14	555	1208	1423.49	89	15.9943	6.9678
7	2	209	470	315.622	15	21.0415	6.8106
8	18	813	1542	3050.08	170	17.9416	7.4887
9	11	392	529	520.99	32	16.2809	7.8859
10	3	166	268	87.0122	4	21.753	5.0802
11	13	424	731	727.813	35	20.7947	7.4846
12	11	235	335	925.352	48	19.2782	6.3267
13	4	325	421	208.238	11	18.9307	7.8808
14	10	360	625	789.713	37	21.3436	8.5944
15	12	411	688	312.078	13	24.006	5.9023
16	3	66	220	153.468	7	21.924	7.6788
17	0	0	0	0	0	0	
18	20	603	1219	2081.06	117	17.7868	8.5339
19	14	432	924	1535.08	74	20.7444	7.8714

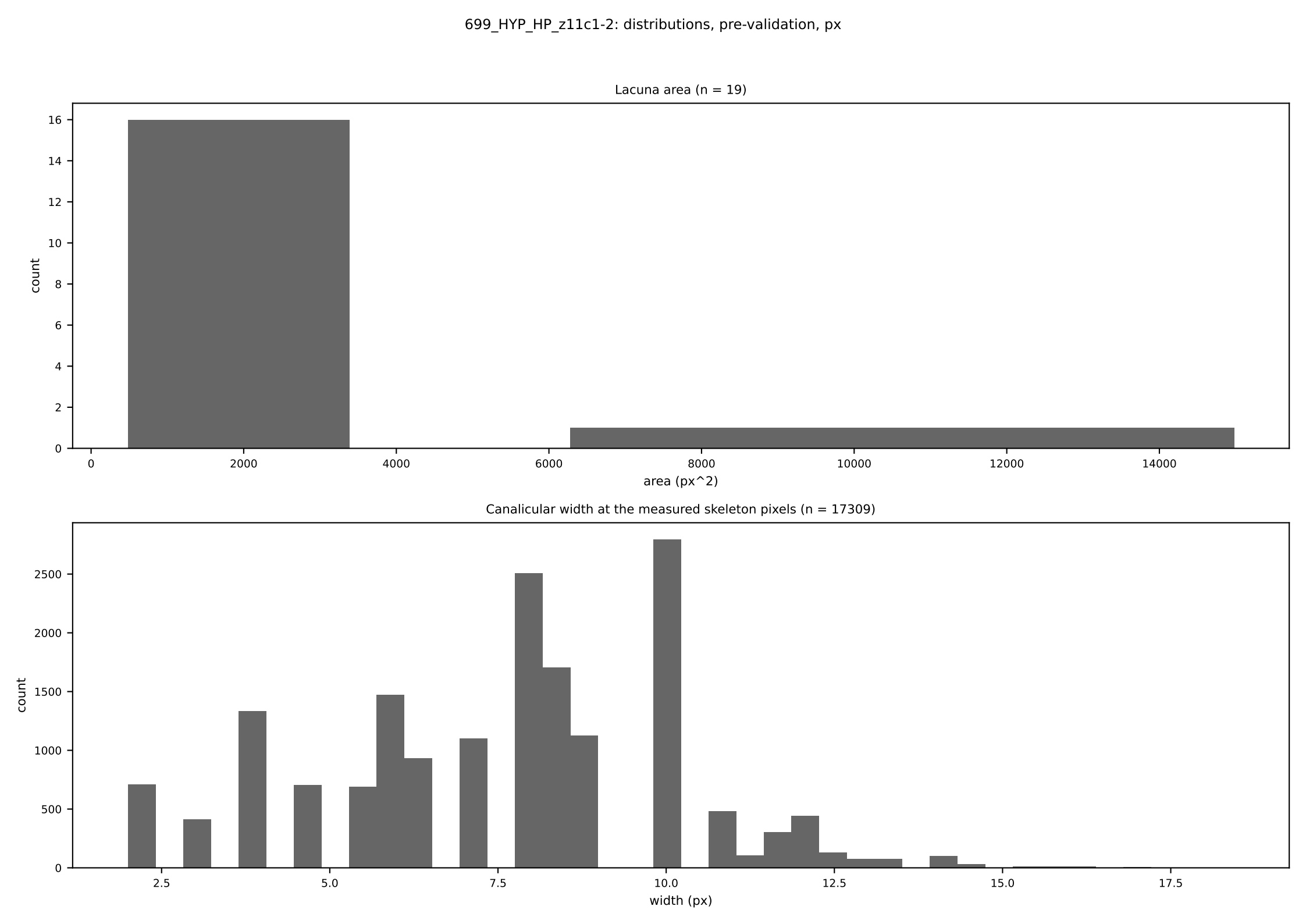
Normalised per lacuna

lacuna_id	perimeter_px	ring_area_r30_px2	ring_area_r60_px2	in_frame_fraction_r30	in_frame_fraction_r60	ring_density_r30	ring_density_r60	roots_per_100px_perimeter
1	95.2548	3348	9123	0.6198	0.5747	0.0340502	0.0279513	3.1494
2	191.622	7458	18845	1	1	0.0505497	0.0387901	6.2623
3	234.25	8068	19697	1	1	0.0478433	0.0353861	5.1227
4	172.995	7102	18424	1	1	0.0425232	0.0319692	4.6244
5	894.833	17281	36859	1	1	0.0513281	0.0455248	2.0115
6	394.333	10570	25661	1	1	0.0525071	0.0470753	3.5503
7	174.066	7351	20270	1	1	0.0284315	0.023187	1.149
8	467.446	11796	28229	1	1	0.0689217	0.0546247	3.8507
9	318.333	9236	16615	1	1	0.0424426	0.0318387	3.4555
10	232.285	6906	15617	1	1	0.0240371	0.0171608	1.2915
11	409.931	10384	22886	1	1	0.0408321	0.0319409	3.1713
12	174.853	5222	9580	1	1	0.0450019	0.0349687	6.291
13	172.267	5412	9650	1	1	0.0600517	0.0436269	2.322
14	182.894	6958	19375	1	1	0.051739	0.0322581	5.4676
15	834.833	23261	48925	1	1	0.0176691	0.0140623	1.4374
16	304.777	8057	16374	0.9164	0.8272	0.00819163	0.0134359	0.9843
17	235.48	4021	7954	0.6927	0.6449	0	0	0
18	1353.43	28693	52942	0.9016	0.8594	0.0210156	0.0230252	1.4777
19	402.333	12473	26315	1	1	0.0346348	0.035113	3.4797

Network v2 per lacuna

lacuna_id	ring_attached_length_r30_px	ring_attached_length_r60_px	ring_length_w_r30_px	ring_length_w_r60_px	sholl_crossings_r10	sholl_crossings_r20	sholl_crossings_r30
1	101	130	122.841	276.823	4	2	2
2	348	593	432.99	829.039	8	8	13
3	374	621	434.049	768.115	9	12	12
4	272	545	349.907	668.215	6	9	7
5	863	1411	993.595	1886.52	25	16	17
6	526	994	621.002	1339	14	12	17
7	169	284	233.752	529.76	4	4	7
8	799	1489	916.37	1721.76	18	18	17
9	333	413	439.948	597.002	10	10	9

10	92	92	186.024	295.38	5	4	4
11	400	491	468.605	820.754	11	13	7
12	207	224	264.309	374.978	5	5	4
13	317	364	377.019	482.446	8	8	4
14	321	490	406.563	701.6	8	11	9
15	246	334	462.688	782.637	8	11	10
16	61	122	72.6985	249.066	1	1	3
17	0	0	0	0	0	0	0
18	438	1131	656.103	1351.76	15	20	15
19	271	473	479.605	1029.15	10	10	17


699_HYP_HP_z11c1-2: distributions, pre-validation, px
Lacuna area (n = 19)
count
area (px^2)
0
2
4
6
8
10
12
14
16
0
2000
4000
6000
8000
10000
12000
14000
Canalicular width at the measured skeleton pixels (n = 17309)
count
width (px)
0
500
1000
1500
2000
2500
2.5
5.0
7.5
10.0
12.5
15.0
17.5